

Phase 2 Report — ACDC Domain-Gap Evaluation

Starting point

The evaluation notebook was first run with the model rebuilt as **DeepLabV3+/ResNet34**, carried over from an early Phase 1 draft, and compared against a hardcoded baseline of **18.73% mIoU** left over from that same draft. Loading the Phase 1 checkpoint into this architecture produced near-uniform, near-zero results across all four adverse-weather conditions: fog 0.32%, rain 0.44%, snow 0.43%, night 0.40%. Diagnosis: the Phase 1 model actually trained and saved was SegFormer/MiT-B4, not DeepLabV3+/ResNet34 — the state dictionary being loaded did not match the architecture being evaluated, so the network was effectively running on incompatible/garbage weights rather than revealing a genuine domain gap.

Rebuilding the Evaluation Setup

Rebuilt the evaluation model to match the architecture actually trained in Phase 1: **SegFormer/MiT-B4**, loaded with **strict** state-dict matching so that any future checkpoint/architecture mismatch fails loudly instead of loading silently. Corrected the hardcoded Cityscapes baseline constant from the stale 18.73% to the true Phase 1 result, **73.18% mIoU (with TTA)**, so the domain-gap comparison would be measured against the correct reference point.

Verifying Checkpoint Integrity

Before re-running on Kaggle, the local copy of **best_model.pth** was inspected directly: 245.8 MB, **61,373,908 parameters**, with state-dict keys beginning **encoder.patch_embed1** through **patch_embed4** — the parameter count matches Phase 1's reported 61,373,395 (the small difference is BatchNorm running statistics) and the key names confirm the SegFormer hierarchical patch-embedding structure, not a ResNet backbone. The checkpoint was then re-uploaded to Kaggle as a dedicated model artifact to remove any ambiguity with the earlier, stale dataset reference.

Missing Ground-Truth Masks Discovered

With the architecture fixed, the first re-run surfaced a second, independent issue: the ACDC dataset attached to the notebook contained the RGB images ("Version 1") but reported **zero ground-truth masks** for every condition and split (fog, rain, snow, night — val and train). Diagnosis: only the image archive had been uploaded; the ground-truth label archive ("Version 2", containing the `gt_labelTrainIds` masks the evaluation code expects) had not been attached at all. Fixed by downloading and attaching the ACDC ground-truth archive, after which mask counts matched image counts exactly for every condition (500 fog, 500 rain, 500 snow, 506 night paired samples).

Final result: 49.02% average mIoU on ACDC

With both issues resolved, the corrected evaluation ran cleanly across all four adverse-weather conditions:

- **Fog:** 60.73% mIoU (−12.45% vs. source)
- **Rain:** 52.37% mIoU (−20.81% vs. source)

- **Snow:** 47.62% mIoU (−25.56% vs. source)
- **Night:** 35.35% mIoU (−37.83% vs. source)

Against the Cityscapes source baseline of **73.18% mIoU**, the average ACDC performance of **49.02% mIoU** represents an absolute drop of **24.16 points** (a **33.0% relative** performance loss) under direct transfer with zero retraining.

The per-condition ordering is consistent with expectations: fog degrades the least, since haze reduces contrast but preserves scene structure and colour cues, while night degrades the most, since the illumination shift is the most severe departure from the daytime-only Cityscapes training distribution. This is a credible, literature-consistent domain gap, ready to serve as the reference point for Phase 3 (DAFormer/MIC unsupervised domain adaptation).

Results — Figures

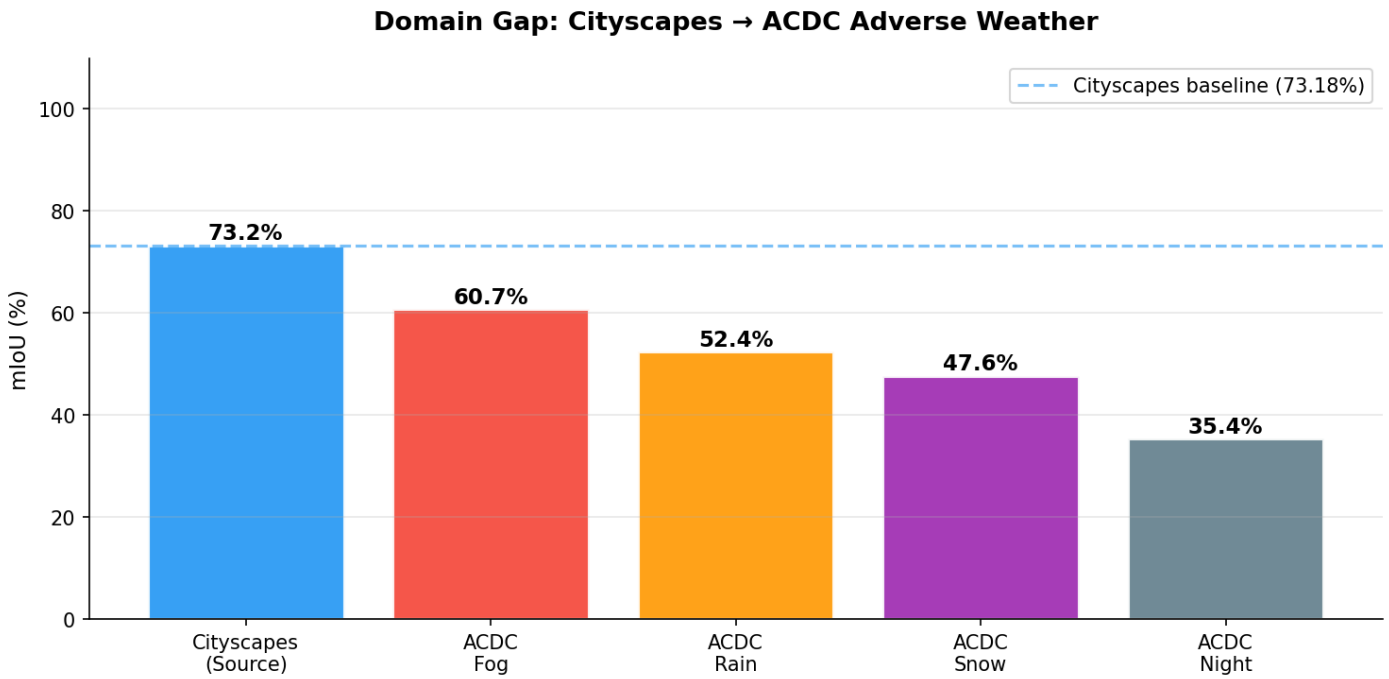


Figure 1: Domain gap from Cityscapes (source, 73.18% mIoU) to each ACDC adverse-weather condition under direct transfer with no domain adaptation.

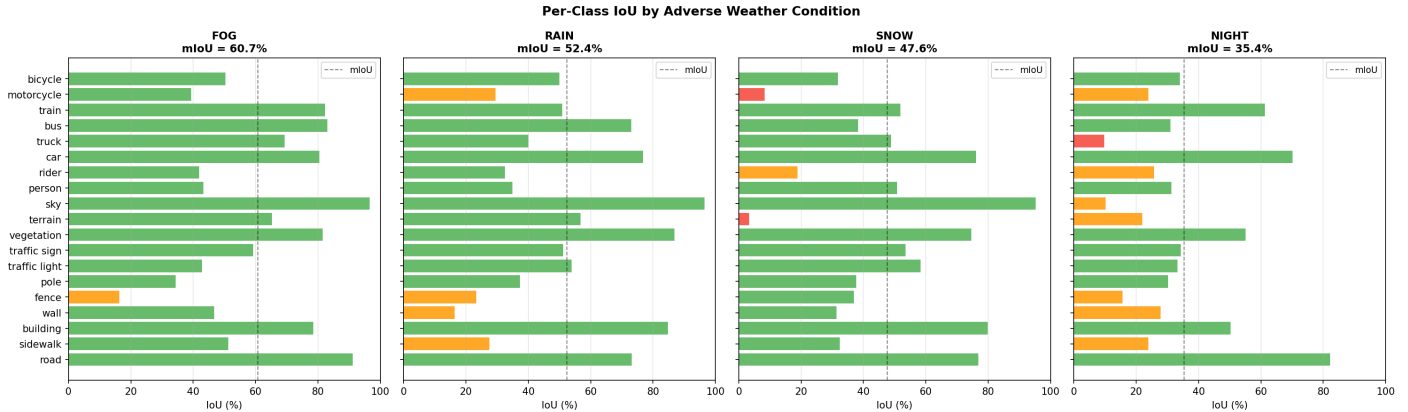


Figure 2: Per-class IoU on ACDC, broken down by condition (fog, rain, snow, night).

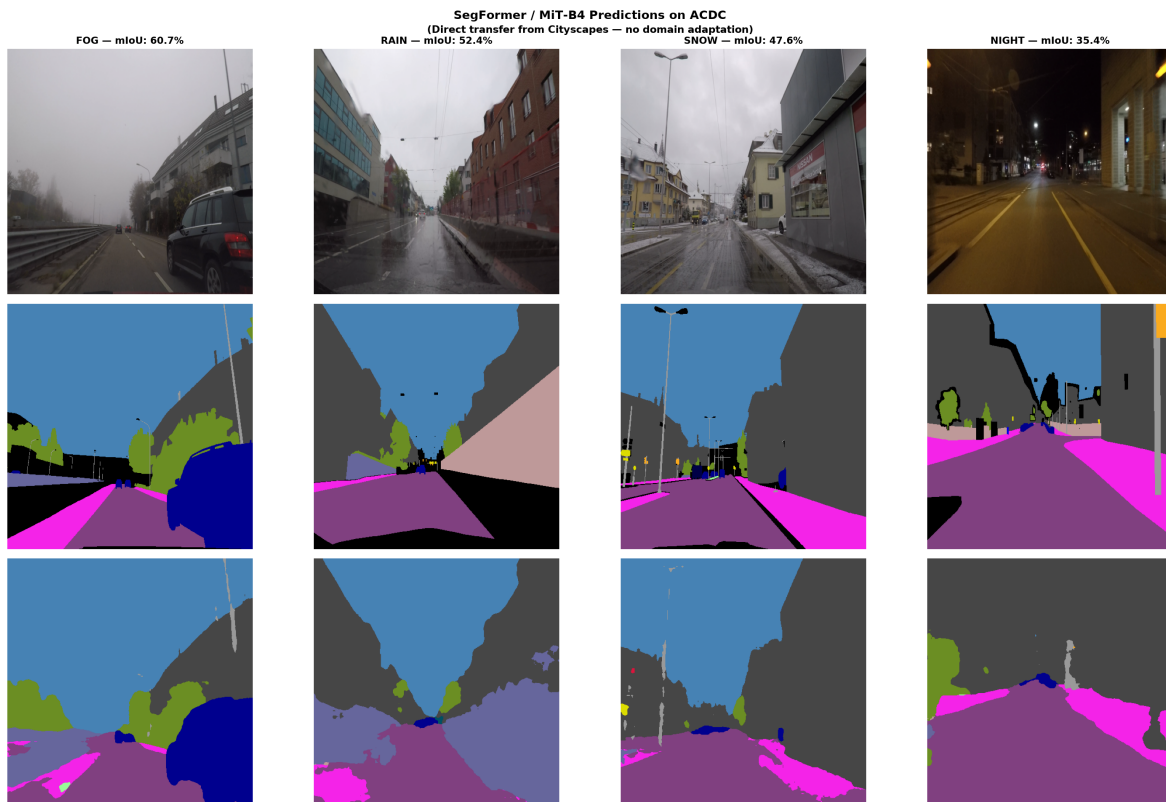


Figure 3: Qualitative SegFormer/MiT-B4 predictions on ACDC under direct transfer (input / ground truth / prediction) for one sample per condition.